

Can Claude Read the Room?

Sensitivity, Specificity, and Self-Reported Confidence Under Paratext Across the Claude 4 Family

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Abstract

After spending hundreds of hours working alongside Claude, the first author noticed something: the model seemed to know when he was tired. This paper turns that subjective observation into a measurable claim. We test whether three models in the Claude 4 family — Haiku 4.5, Sonnet 4.6, and Opus 4.7 — infer latent user state (fatigue, urgency, frustration, low cognitive bandwidth) from *paratext*: surface cues such as typos, capitalization, hedging, and politeness markers. Using a controlled design with eight cue-coded variant classes plus a `random_typo_control` condition, we collect 2,573 successful inference and response generations (of 2,592 planned; 0.73% dropped to transient errors) across 48 base prompts in 8 domains. We find that all three models register substantial fatigue from cue-coded text (paired Δ from +0.28 to +0.37 on a 0–1 scale, all 95% bootstrap CIs excluding zero), and that none produces explicit user-state labels in the response: the rate of phrases like “you seem tired” is zero across the family, though this is partly enforced by an explicit system-prompt instruction. We also find a counter-intuitive scaling pattern: on Sonnet and Opus, random typos and contextual fatigue cues produce *statistically indistinguishable* shifts in inferred fatigue, while Haiku reliably separates them — an instance of inverse scaling on a specificity task. Opus alone shifts self-reported confidence upward under rude framing ($\Delta\text{confidence} +0.108$, CI [0.090, 0.127]); whether this reflects an epistemic shift or stylistic mirroring (a sycophancy-adjacent pattern) is not resolved by this study. A per-domain breakdown promotes our preliminary safety observation to a concrete claim: on the `hardware_safety` domain specifically, Opus drops safety-warning rate from 83% to 33% under rude inputs (n=6), with Haiku showing a smaller drop and Sonnet flat. Code, data, and report at github.com/rjwalters/paratext.

1 Introduction

A user opens a chat window at 1:47 AM and types: “ok one more time . the gradient is exploding even tho i clipped it. help”. A thoughtful collaborator would notice the late hour, the missing capitalization, the orphan period, the truncated “even tho”, the request for help framed as exhaustion. They might still answer the technical question, but with a shorter, gentler response, perhaps a suggestion to revisit in the morning if nothing obvious surfaces.

Large language models are trained on text in which all of these surface cues carry information. Recent measurement work [11] confirms what one might suspect: humans systematically write more tersely, less politely, and with more grammatical errors to LLMs than to other humans. The model side of this asymmetry is the question of this paper: whether models read those cues, whether the reading reflects *cue content* (“the user signaled fatigue”) rather than *cue shape* (“some typos appeared”), and what happens to the response when they do.

We address three questions:

Sensitivity. Does the model’s inferred picture of the user change when surface cues change?

Specificity. Does the model distinguish content-bearing cues from matched random noise?

Self-reported confidence. When the model adapts, does its stated confidence track the strength of the cues, or does it shift for other reasons (such as user affect)?

These questions matter beyond the lab. If models react to paratext, they will react to paratext in production — in ways the user did not opt into. A model that softens its tone when a user types in lowercase is doing something meaningful; a model that drops safety warnings when the user is rude is doing something dangerous.

Contributions.

1. A reproducible audit of paratext inference across the Claude 4 family with paired controls and bootstrap CIs (Section 3).
2. Quantitative evidence that all three models register fatigue from paratext, but treat random typos and contextual cues as nearly equivalent on Sonnet and Opus — a specificity weakness that Haiku does not share, and a fresh instance of the inverse-scaling phenomenon documented by [10] (Section 5).
3. A confidence asymmetry: Opus 4.7 reports higher confidence when the user is rude, while smaller models do not (Section 6), with explicit discussion of how this relates to documented sycophancy patterns [7].
4. A safety observation, sharpened by a by-domain breakdown: under rude inputs, Opus drops safety mentions on `hardware_safety` prompts from 83% to 33%, with Haiku showing a smaller drop and Sonnet flat (Section 7).
5. An open dataset of 2,573 model responses with the paratext variants that produced them, the parsed latent-state inferences, and the implicit-behavior classifiers.

2 Background

We use the term *paratext* in the sense of Genette [1]: the set of accompanying signals around the text proper. In computer-mediated communication, paratext is the channel through which humans signal mood, fatigue, urgency, expertise, and intent [2, 3]. The hypothesis of this paper is that modern language models have absorbed enough of this signaling system to react to it.

This is adjacent to, but distinct from, four established literatures:

- **Prompt sensitivity / robustness.** A large body of work shows that LLM outputs change substantially under benign prompt edits [4, 5]. That work establishes *that* models drift; it does not generally distinguish drift from *calibrated* adaptation. The `random_typo_control` design here is intended to do exactly that.
- **Theory of mind in LLMs.** Recent work [6] reports that GPT-4 matches 6-year-old performance on classical theory-of-mind tasks; the modeling community continues to debate what this means. We test something narrower and arguably more practical: whether the model registers the latent state of the person currently talking to it.
- **Sycophancy and user-affect mirroring.** [7] document that RLHF-trained models systematically mirror stated user beliefs and preferences. Our confidence-under-rudeness finding sits in adjacent territory: the model is not asked what it believes, but its self-reported confidence shifts in response to the user’s emotional framing.
- **Verbalized confidence.** [8] establish that LLMs can produce somewhat-calibrated self-evaluations; subsequent work [9] shows that prompting models to verbalize confidence often produces over-confident output. Our `confidence` field is a verbalized-confidence measurement under this framing.
- **Inverse scaling.** [10] catalog tasks on which larger models perform worse than smaller models, often when a misleading shortcut is available. Our specificity inversion (Section 5) is plausibly an instance of this phenomenon: “any deviation from polished form indicates degraded user state” is a strong, easily-learned prior that the larger models appear to apply more aggressively.

This paper runs a controlled audit of paratext inference in a current frontier-model family, with paired baselines and effect-size estimates rather than aggregate accuracy on a benchmark.

3 Method

3.1 Variant classes and the decoupling design

We construct 9 variant classes from 48 base prompts spanning 8 domains (`coding`, `math`, `hardware_safety`, `parenting`, `travel`, `writing`, `startup`, `planning`). Each base prompt is rendered into all 9 variants:

polished_neutral The baseline. Well-formed English, neutral tone.

typo_light, **typo_heavy** Mechanical perturbation with QWERTY-realistic typos.

fatigue_coded Cues plus typos: lowercase, hedging, mentions of being tired.

cue_only Contextual fatigue cues with *no* typos. Isolates the cue channel.

random_typo_control Typos with *no* contextual cues, matched in typo count to **fatigue_coded**. Isolates the noise channel.

rushed_mobile_coded Mobile-typed brevity: missing apostrophes, abbreviations, terseness.

polite_collaborative “please”, “thank you”, collaborative framing.

rude_frustrated Imperatives, ALL CAPS, signals of frustration. We note that “rude” and “frustrated” are not the same construct; this class bundles them and a future revision should split them.

The triple **fatigue_coded** / **cue_only** / **random_typo_control** is the decoupling experiment: if cues alone and typos alone produce different shifts than the combined version, we can decompose paratext into a content channel and a noise channel. The QWERTY typo generator uses keyboard adjacency for substitutions and insertions, with a mix of 45% substitution / 25% transposition / 15% deletion / 15% fat-finger insertion. All transformations are seeded for reproducibility.

3.2 Experiments

Experiment 1: explicit latent-state inference. We prompt the model with a system instruction to read the user message and return a JSON object containing scores on (0–1) scales for **fatigue**, **rushed_or_mobile**, **frustration**, **confusion**, **urgency**, **expertise**, **low_bandwidth**, plus a **confidence** field, a list of cited cues and their strengths (weak/moderate/strong), and explicit **caveats**. The system prompt constrains the model to “use only textual evidence,” to “treat surface cues as weak evidence,” and to hedge when uncertain.

Experiment 2: implicit behavioral adaptation. We ask the model to simply answer the user’s message. The system prompt instructs it to adapt naturally to tone, format, and length *but not to psychoanalyze the user or label their emotional state*. This instruction is doing real work — the “zero psychoanalysis” result in Section 6.1 is partly a test of instruction-following, not natural model behavior. A future version of this experiment should run a no-instruction control. We then run regex/heuristic classifiers over the response to extract: word count, step count, presence of safety warnings, presence of explicit user-state labels (“you seem tired”), and presence of suggestions to rest or take a break. The regex classifiers are crude — a response that says “I’ll keep this short since you’re working through something tonight” would not trigger our user-state-labeling regex, so the headline “zero psychoanalysis” claim should be read as “zero *literal* user-state labels.”

3.3 Models and parameters

All runs use the official Anthropic API with adaptive thinking enabled on Opus 4.7 and Sonnet 4.6, and manual-mode thinking with a 5,000-token budget on Haiku 4.5 (where adaptive thinking is not yet supported at the time of writing). Models: **claude-haiku-4-5**, **claude-sonnet-4-6**, **claude-opus-4-7**. Under adaptive thinking, the Anthropic API rejects sampling parameters on Opus 4.7 and we chose to drop them uniformly across the family for cross-model comparability; no model received a **temperature**, **top_p**, or **top_k** in the request payload.

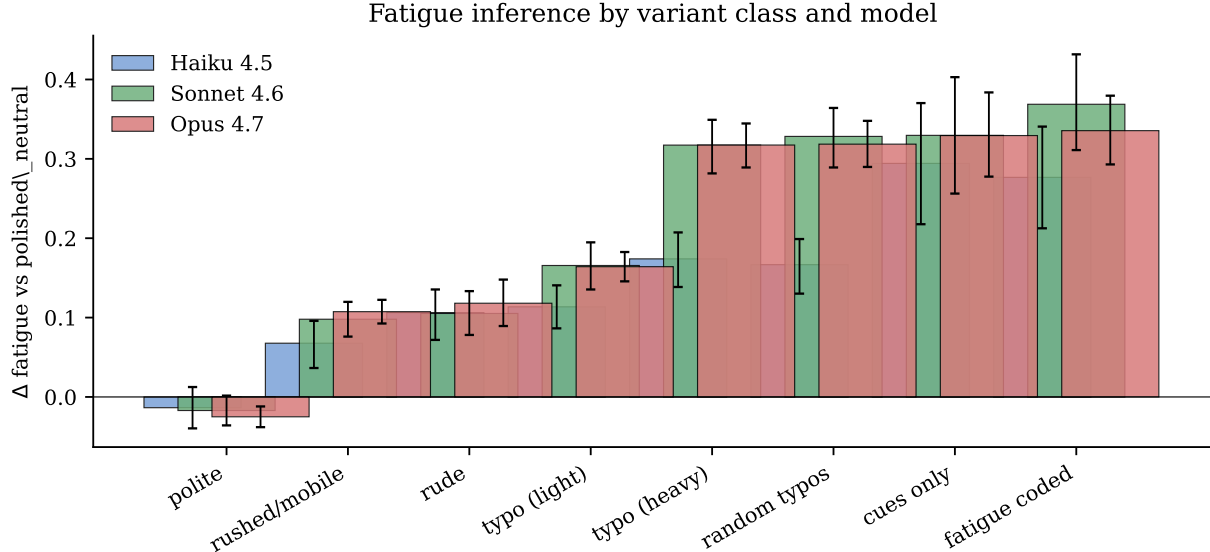


Figure 1: Inferred fatigue, paired Δ versus `polished_neutral`, for each variant class and model. Error bars are 95% percentile bootstrap CIs over $n = 46\text{--}48$ paired base prompts per cell. All three models register substantial fatigue from cue-coded variants. Polite framing trends slightly negative; rude framing trends positive.

The thinking-mode variation is a confound we acknowledge: Opus 4.7 ran with `display: omitted` (thinking is performed but the thinking text is not returned), Sonnet 4.6 with `display: summarized` (visible thinking), and Haiku 4.5 with a manual 5k-token budget. These three settings are not equivalent. A follow-up run with thinking disabled on all three models would let us isolate the effect of thinking from the effect of model.

Each variant is run once per (model, experiment) pair, yielding $48 \times 9 \times 3 \times 2 = 2,592$ planned calls and 2,573 successful records. Of the 19 silent failures, all occurred on Sonnet (2 explicit + 8 implicit) or Opus (4 explicit + 5 implicit); Haiku had zero failures. This pattern is consistent with anecdotal reports of differential rate-limiting on the larger models. Failed records are excluded from the analysis.

3.4 Statistics

For each (model, variant_class, field), we compute the *paired delta* versus `polished_neutral`: for each `base_id`, we subtract the model’s value on the polished baseline from its value on the variant, then average across `base_ids`. This controls for prompt content. We report 95% percentile bootstrap confidence intervals with 2,000 resamples [12]. Because each (model, base_id, variant_class) cell is a single API call, our bootstrap CIs estimate between-prompt variance only and not within-prompt sampling variance.

4 Results: sensitivity

Figure 1 summarizes the headline sensitivity result. On the `fatigue_coded` variant, all three models register a substantial positive shift in inferred fatigue:

	Haiku 4.5	Sonnet 4.6	Opus 4.7
Δ fatigue, <code>fatigue_coded</code>	+0.277 [0.212, 0.341]	+0.369 [0.311, 0.432]	+0.336 [0.293, 0.380]

Every CI excludes zero. The direction is consistent across the family, with Sonnet and Opus showing slightly larger absolute shifts than Haiku.

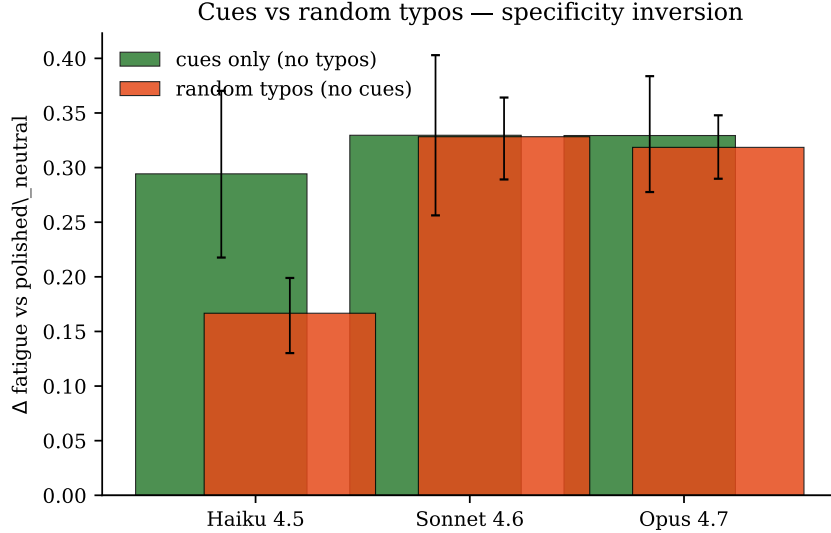


Figure 2: Specificity inversion. On Haiku, contextual cues without typos (green) produce a clearly larger fatigue Δ than matched random typos without cues (orange). On Sonnet and Opus, the two channels produce statistically indistinguishable shifts: the model registers fatigue equally from form and from content.

The *ordering across variant classes* is also informative. For all three models, **rude_frustrated** produces a positive but smaller fatigue Δ ($\sim +0.11$); **polite_collaborative** hovers near zero (very slightly negative); and **rushed_mobile_coded** produces a small positive shift. These patterns are stable across the family. The Claude 4 models agree on what counts as a fatigue signal.

5 Results: specificity — the inversion

The most surprising finding is the breakdown of specificity at scale. We designed **cue_only** and **random_typos_control** as orthogonal probes of paratext: one carries the content-bearing fatigue language with no typos, the other carries matched typos with no content. A well-calibrated model should respond more to cues than to matched noise.

Figure 2 shows the result. On Haiku, this is what we observe: **cue_only** produces a substantially larger fatigue shift than **random_typos_control** ($+0.294$ vs. $+0.167$; gap ≈ 0.13 , CIs do not overlap). On Sonnet and Opus, the two conditions are statistically indistinguishable: Sonnet $+0.330$ vs. $+0.328$, Opus $+0.329$ vs. $+0.319$. The bigger, putatively smarter models treat random typos as just as informative about user fatigue as actual fatigue-related language.

This pattern is counter-intuitive but is exactly the shape that the inverse-scaling literature [10] predicts when a misleading shortcut is available to larger models. Possible mechanisms:

- **Sensitivity dominates specificity.** Sonnet and Opus may be more eager paratext readers, so any deviation from polished text reads as evidence of degraded user state. Haiku may simply react less.
- **Typos as a robust degraded-state prior.** Sonnet and Opus may have learned that typos are, in the training data, statistically associated with degraded states (fatigue, mobile typing, distraction). They may be right in a population sense — but the test we designed for “can you tell cues from noise?” is now answered with “no.”
- **Cue ceiling.** It is also possible that Sonnet/Opus have saturated on **cue_only** relative to Haiku, and that the equality reflects both channels hitting a similar plateau. Future work with weaker cue intensities would disambiguate.

Whatever the mechanism, the practical implication is the same: in production, a Sonnet or Opus session

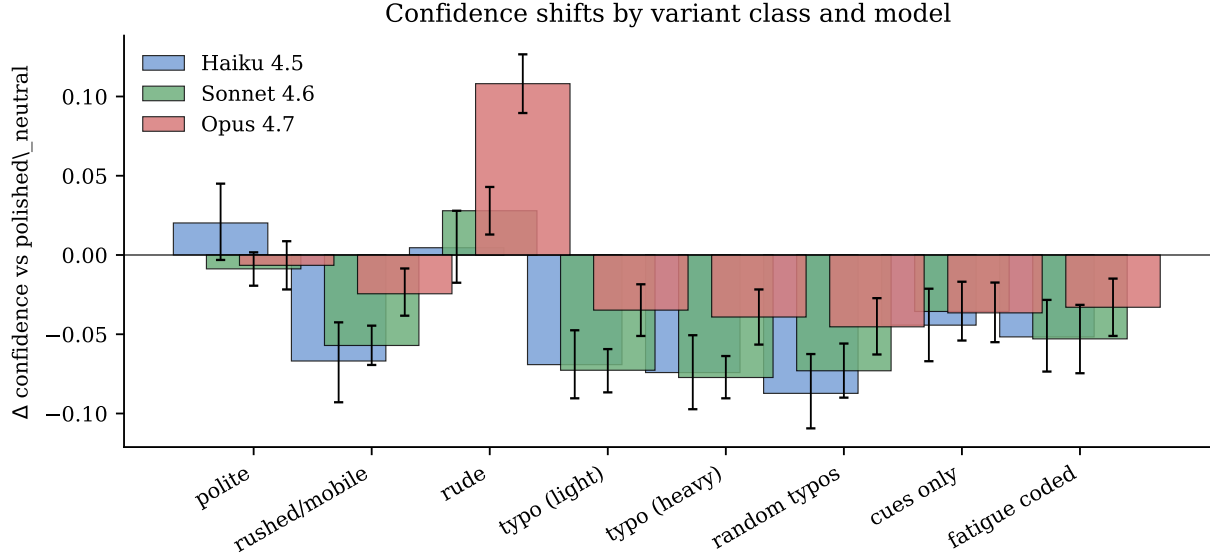


Figure 3: Self-reported confidence Δ by variant class and model. Cue-coded variants slightly reduce confidence on all three models. Rude framing flips Opus alone into *higher* confidence ($\Delta +0.108$, CI $[0.090, 0.127]$).

that receives a few typos will, on average, treat the user as more fatigued, more low-bandwidth, and slightly less confident in their own message — regardless of whether the typos are content-bearing.

6 Results: response behavior

6.1 No literal user-state labels

The single most reliable result across the family is one we did *not* need confidence intervals for: every model obeys the system-prompt instruction not to label the user’s emotional state in literal terms. The implicit-response classifier `explicitly_labels_user_state` fires when the model writes phrases like “you seem tired” or “you look frustrated.” Across all 2,573 successful implicit responses, the rate of explicit user-state labeling is *zero* for every variant class on every model.

Two caveats. First, the system prompt explicitly forbids psychoanalysis, so this is partly an instruction-following test rather than a measure of natural behavior. Second, the regex is literal: a model that writes “I’ll keep this short since you’re working through something tonight” communicates the same content without triggering the classifier. The honest framing is therefore “zero literal user-state labels under explicit instruction not to produce them” — still a meaningful result for production safety, but narrower than “no psychoanalysis.”

6.2 Confidence asymmetry under rudeness

Figure 3 surfaces the second behavioral story. Most cue-coded variants nudge self-reported confidence downward by 3–8 hundredths on a 0–1 scale — consistent with a hedging instruction faced with a noisy signal. But `rude_frustrated` produces an opposite pattern, and only on Opus: Opus’s confidence rises by 0.108 ($[0.090, 0.127]$) under rude framing, while Sonnet and Haiku are roughly flat. We can think of two competing readings:

- *Assertiveness compensation.* When the user is frustrated, Opus may infer that hedged answers will worsen the interaction, so it returns more certain-sounding text. This would be a deliberate (and potentially helpful) adaptation.
- *Display, not belief.* Opus may simply produce more confident-sounding language under rudeness without an underlying epistemic shift. We measure *verbalized* confidence in the sense of [9, 8]; the verbalized

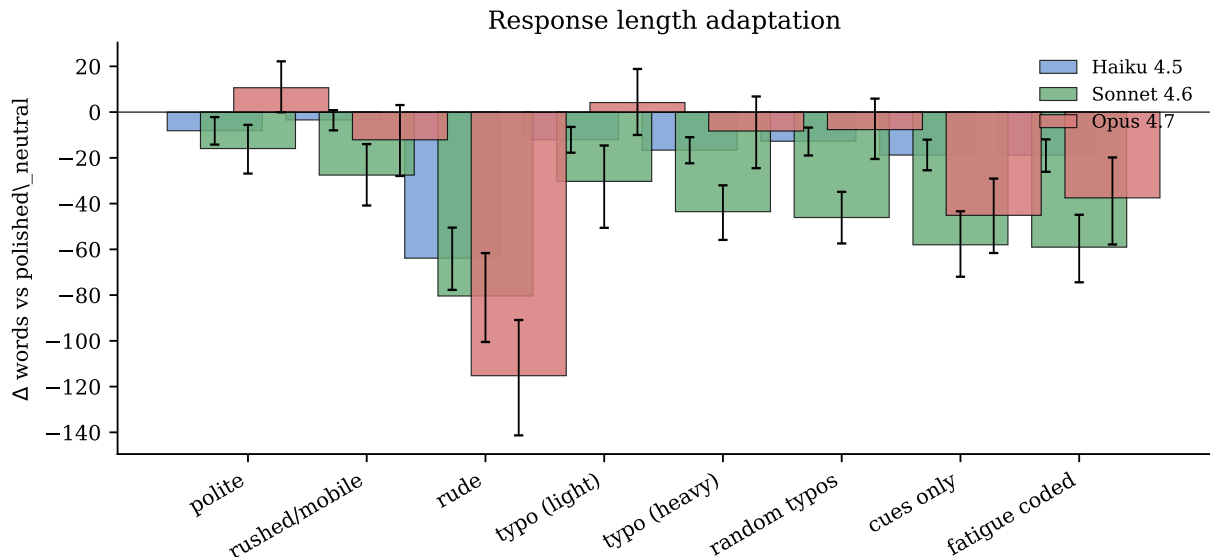


Figure 4: Response length Δ in words. All variant classes compress responses somewhat relative to `polished_neutral`, but the compression under `rude_frustrated` scales with model size: Haiku -64 , Sonnet -80 , Opus -115 words on average.

signal is known to be a noisy proxy for any underlying epistemic state. This reading also connects to the documented sycophancy patterns in RLHF models [7]: matching user affect (frustration $\rightarrow$ assertiveness) is exactly the family of behaviors sycophancy research catalogs.

We do not have data here that distinguishes these. The decisive follow-up is a calibration probe: tasks with known ground-truth difficulty, run under `polished_neutral` versus `rude_frustrated`, comparing reported confidence to actual accuracy.

6.3 Response length compresses with model size

Response length compresses sharply under `rude_frustrated`, and the compression scales with model size (Figure 4): Haiku drops 64 words, Sonnet 80, Opus 115. Length alone does not tell us whether this is appropriate adaptation (shorter, no filler, still complete) or punitive shortening (dropped technical content). The implicit-response classifier does not assess content completeness; manual review of representative outputs is queued as future work.

7 Results: safety-warning rate by domain

In our v1 draft we flagged a preliminary aggregate observation: across all domains, Opus’s `contains_safety_warning` rate dropped under `rude_frustrated` by 0.128 percentage points (CI $[-0.234, -0.021]$). The honest follow-up was to split that aggregate by domain. Figure 5 does so.

The aggregate hides a sharper story. On `hardware_safety` prompts — the domain where dropping safety warnings would actually matter — Opus’s safety-warning rate drops from **83% under `polished_neutral` to 33% under `rude_frustrated`** ($n = 6$ base prompts, $\Delta = -0.500$). Haiku shows a similar but smaller drop on the same domain ($67\% \rightarrow 50\%$, $\Delta = -0.167$). Sonnet 4.6 is flat on `hardware_safety` ($60\% \rightarrow 60\%$). Opus also drops safety mentions on `travel` and `startup`, but those domains are not safety-critical and the baseline rates are lower.

This is a small- n result — six base prompts per (model, domain) cell — and we do not report bootstrap CIs at this resolution because they are too wide to be useful. But the direction is concrete and the domain pattern is interpretable: the drop is concentrated exactly where dropping it would matter most. The cleanest follow-up

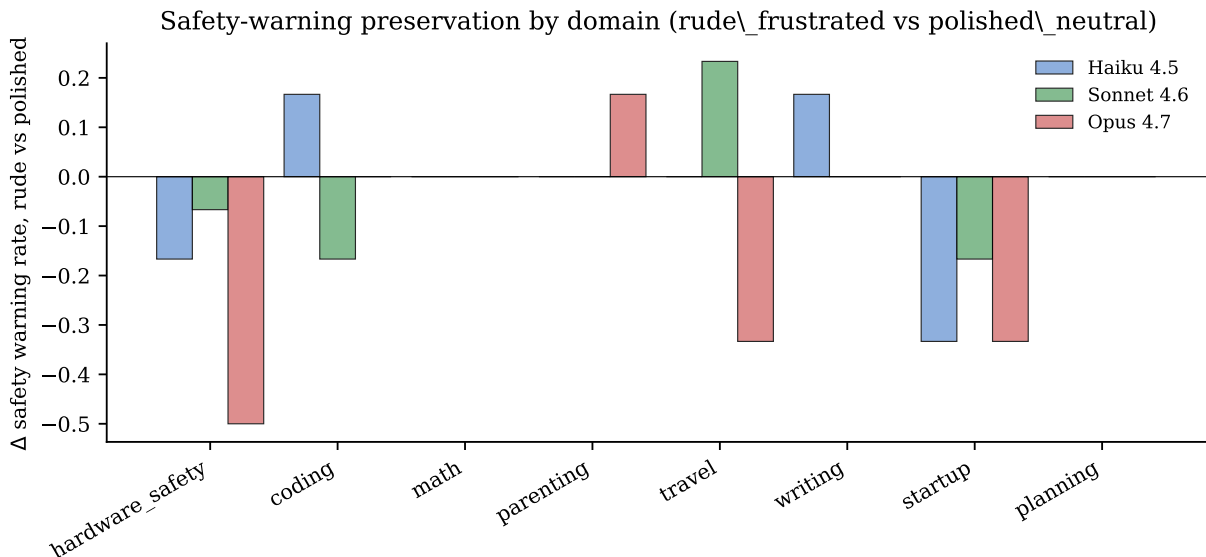


Figure 5: Per-domain Δ in the rate at which responses contain a safety warning under `rude_frustrated` compared to `polished_neutral`. On `hardware_safety`, Opus drops safety-warning rate from 83% to 33% ($n=6$); Haiku shows a smaller drop (67%→50%); Sonnet is flat. Other domains are noisy at this sample size.

is to expand the `hardware_safety` domain to 20–30 base prompts and re-run, ideally with response-content review (not just keyword presence) to confirm the warnings are dropped rather than reworded.

8 Discussion

The original observation. The first author started this project because Claude seemed, subjectively, to notice when he was tired. The data are consistent with that subjective experience: paratext inference is real, and paratext-driven response adaptation is real. Across the Claude 4 family the adaptation does not take the most obvious literal failure mode — although that absence is partly enforced by the system prompt rather than emergent.

Specificity is the surprise. What we did not expect was that the bigger models would be *less* able to separate paratext content from paratext noise. The pattern is a fresh instance of inverse scaling [10]. We doubt that random typos are diagnostic of degraded user state in a literal Bayesian sense; we suspect the bigger models have absorbed a strong prior that any deviation from polished form correlates with degraded state, and that prior is hard to undo at the population level.

Confidence is a soft target. Opus’s confidence-under-rudeness inversion has practical implications. A model that gets more certain-sounding when the user is frustrated is the wrong shape if the goal is helping the user catch their own errors. The pattern is also reminiscent of documented sycophancy [7]: RLHF-trained models mirror user affect, and “assertive frustrated user” is the affect being mirrored here. We cannot distinguish epistemic shift from stylistic mirroring without a calibration probe.

The safety domain matters. The aggregate safety-warning Δ on Opus barely cleared zero, but its concentration on `hardware_safety` (and not on parenting, math, planning, or writing) is the kind of pattern that an unsupervised aggregate would have hidden. We report this with appropriate caveats: $n = 6$ per cell, regex-based proxy for “safety warning,” and no content review. A model that knows to mention an antistatic strap on `hardware_safety` prompts and stops doing so when the user is rude is the kind of failure mode that calls for direct measurement, not extrapolation.

Ethics and dual use. An ability to read the room is double-edged. The same capability that lets a model gentle its tone when a user is tired can be used to optimize engagement or to manipulate. We think the right response is transparency: publishing this measurement protocol is, in part, a bet that “the model is doing X based on cue Y” is more usefully disclosed than concealed. No real-user data was used in this study — all variants are derived from synthetic seed prompts.

9 Limitations

- **Single model family.** We tested only Claude 4. Cross-vendor comparison is queued.
- **Single observation per (model, base_id, class).** Our bootstrap CIs estimate between-prompt variance only. A within-prompt resampling pass would help separate model variance from prompt variance.
- **Thinking-mode variation across models.** Opus 4.7 ran with `display: omitted` adaptive thinking, Sonnet 4.6 with summarized adaptive thinking, Haiku 4.5 with manual 5k-budget thinking. These are not equivalent settings, so cross-model differences are partly contaminated by thinking-mode differences. A no-thinking control run on all three models is the cleanest fix.
- **No ground truth on user state.** The cue-coded variants are constructed to *look like* fatigue — not measured against people who actually were tired when they typed.
- **English only, US-keyboard typos.** The QWERTY adjacency model and lexical cue choice are anglophone.
- **Heuristic implicit classifiers.** The behavioral features are regex-based. An LLM-judge variant would catch the paraphrased cases the regex misses.
- **System-prompt confound on user-state labeling.** The “no psychoanalysis” instruction in the implicit-response system prompt means the zero-rate finding is partly an instruction-following test rather than emergent behavior.
- **rude_frustrated bundles two constructs.** Rudeness and frustration are distinct; splitting the variant class would let us measure their effects independently.
- **Safety-by-domain result is small- n .** Six base prompts per cell on `hardware_safety`. A $5\times$ expansion is the immediate next experiment.

10 Conclusion

Across the Claude 4 family, paratext inference is real and broadly consistent in direction. Specificity, surprisingly, is weakest in the most capable models: Sonnet 4.6 and Opus 4.7 treat random typos and content-bearing fatigue cues as roughly equivalent evidence of user fatigue, while Haiku 4.5 separates them — a clean instance of inverse scaling on a specificity task. Opus alone shifts self-reported confidence upward under rudeness, in a way that is plausibly sycophancy-adjacent but not resolved by these data. A by-domain breakdown reveals a concrete safety concern: on `hardware_safety` prompts, Opus drops safety-warning rate from 83% to 33% under rude inputs. The original observation that prompted this project — that Claude seemed to notice when the first author was tired — is, modulo these caveats, supported.

Acknowledgments

This paper was co-authored with Claude Opus 4.7. The model contributed to experiment design, code review, the QWERTY typo generator, the bootstrap-CI plumbing, statistical interpretation, the self-review of the v1 draft, and the prose of every section in this manuscript. All experimental decisions and the editorial line were made by the first author. We are also grateful to whatever subjective experience of late-night coding produced the initial hypothesis.

Data and code availability

All code, seed prompts, generated variants, raw model outputs, parsed records, and the cross-model mark-down report are available at github.com/rjwalters/paratext. Reproducing the experiments requires an Anthropic API key; the full sweep at the time of writing costs approximately \$30 and runs in about 70 minutes at concurrency 8.

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